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# Assignment 1

Que1 List the student name, dob from table2

Answer Select student name, dob from student.

Que2 List name of students scoring more than 3.7 in GPA?

Answer Select sname from students where GPA > 3.7.

Q3 List the name of students whose high school size is atleast 1000 and born after 1996.

Ans Select sname from students where size >=1000 and dob > 12/31/1996.

Q4 List the name of students who are scoring GPA in between 2.9 and 3.9.

Ans select Sname from student where GPA > 2.9 and GPA < 3.9

Q5 List all the details of college who situated in MA.

Ans Select * from college where State = MA.

Q6 List the student who are scored more than 2.0 but less than 3.5

Ans Select Sname from students where GPA > 2.0 and GPA < 3.5

Q7 List the students who have born after 1st Jul 96 in the order of the date of birth.

Select Sname from students where DOB > '06/01/1996'

Q8 List the SID, CNAME, decision of application are accepted.

Select SID CName from Apples where decision = 'Y';

Q9 List the SID CName of applications which are filled on Stanford.

Select SID Name from Apply where Name = 'Stanford'

Q10 List the colleges that has enrollment greater than 10001

Select * from colleges where enrollment > 10001

Q11 List the colleges not in california

select * from college where state != california.

Q12 List the name of all student who from high school having size greater than 17000 and scored GPA less than 3.8.

Select sname from students where sizeHS > 17000 and GPA < 3.8

Q13 Display the disription of student table.

Desc student

Q14 Display the details of all students

select * from student

Q15 Display unique major.

Select Distinct * from Student.

Q16 List the student names those are having three character in their names.

Select Sname from Student where Sname like '___'

Q17 List the student names those are starting with H and with five character.

select sname from students where sname like 'H%' and sname like '_____';

Q18 List the student names those are having third character and forth char must be 'e'.

Select * from students whose sname like '__e_e';

Q19 List the student name ending with 'y'

Select sname from student whose sname like '%y';

Q20 List the students in the order of their CPA.

Select sname, CPA from student order by CPA asc;

Q21 List the details of the students in order of the ascending of CPA and descending of DOB.

Select * from student order by CPA asc, DOB desc;

Q22 List the SIDs of student who apply in either 'Standford', 'cornell' or 'MIT' college.

Select SIDs from student where apply = 'Standford' or 'Cornell' or 'MIT';

Q23 Delete all applications filled at standford.

Delete from apply where Name = 'Standford';

Q24 Delete the college Standford from college table.

Delete from college where Name = 'Standford';

Q25 Modify the CPA of all students by giving 10% raise in their CPA.

~~Modify the CPA by giving 10% raise~~

select *gpa (0.1*gpa) as final from student;

" Create the following

Q26 Increment the GPA of the students by 1.5 whose GPA is less than 3.5 and belong to High School having size greater than 1500

```
Select * from student set GPA = GPA + 1.5 where
    SIZE HS > 1500;
```

Q27 Delete the students whose GPA is less than 3.2

```
Delete from student where GPA < 3.2;
```

# Assignment-2

1. Create the following tables and specify constraints at the time of creation.

(i) Department

| Column Name | Data Type | Size | Constraint |
|---|---|---|---|
| Deptno | number | 3 | primary key |
| Dname | varchar2 | 20 | Unique |
| Location | varchar2 | 20 | not null, department are located in Delhi, Pune, Agra |

Sol-(i) 1. create table Department (Deptno number(3) primary key, Dname varchar2(20) unique, Location varchar2(20) Not null check (location in ('Delhi', 'Pune', 'Agra')));

2. Desc Department;

(ii) Employee

| Column name | Data Type | Size | Constraints |
|---|---|---|---|
| Empno | varchar2 | 5 | primary key, should start with 'E' |
| Ename | varchar2 | 20 | Unique |
| Designation | varchar2 | 20 | not null |
| Salary | number | 10 | default 25000, must lie between 15000 and 50000 |
| DOB | date | | not null |
| Dno | number | 3 | foreign key (references department) |

Sol (ii)

```
CREATE table Employee
(Empno varchar2(5) primary key check(Empno like 'E%'),
Ename varchar2(20) unique,
Designation varchar2(20) not null,
Salary number(10) default 25000
check(salary between 15000 and 50000),
Dob date not null,
Dno number(3) references Department);
desc Employee
```

(iii) Candidate

| Column Name | Data type | Size | Constraints |
|---|---|---|---|
| Candidate_ID | Number | 6 | Primary key of the table |
| Candidate_Name | varchar2 | 20 | Not Null |
| Candidate_Email | varchar2 | 30 | Unique Must have '@' followed by '.' in between the email |
| Candidate_Dept | Number | 2 | Default 'HR' |
| Manager_ID | varchar2 | 30 | It can take only those values which are present in candidate_ID column |

Soln

```
CREATE table Candidate
(Candidate_ID number(6) primary key,
candidate_name varchar2(20) Not null,
Candidate_Email varchar2(30) unique.
check (Candidate_Email like '%@%.%'),
Candidate_Dept varchar2(2) default 'HR',
Manager_ID integer references Candidate(Candidate_ID)
);
desc Candidate;
```

2. Create the schemas as specified above without specifying any constraints.

College (cName : varchar2(10), state : varchar2(10), enrollment : int)
Student (sID : int ; sName : varchar2(10), GPA : number(2,1), sizeHS : int)
Apply (sID : int ; cName : varchar2(10), major : varchar2(20))

Sol 1. CREATE TABLE College (cName varchar2(10), state varchar2(10), enrollment int);

2. CREATE TABLE Student (sID int, sName varchar2(10), GPA number(2,1), sizeHS int);

3. CREATE TABLE Apply (sID int, cName varchar2(10), major varchar2(20));

4. DESC college;
5. DESC Student
6. DESC Apply;

(i) Add cName as Primary key in College

Ans Alter table college add primary key (cName);

(ii) Add sID as Primary key in student

Ans Alter table student Add primary key(sID);

(iii) Add sID, cName, major as Primary key in Apply.

Ans alter table apply Add primary key (sID, cName, major);

(iv) Make sID in Apply foreign referring table student and cName referring table college

alter table apply add constraints fk_student foreign key (sID) references student;

(v) Increase data type size of majoration colors.

11. Ans alter table apply modify(majoration varchar2(3));

(vi) Add a new column decision in the Apply table keeping a constraint of not null for this column with data type varchar2(3)

Sol alter table apply add(decision varchar2(3) not null);

(viii) Drop foreign key on column name cName from Apply table.

Ans alter table apply drop constraints fk_cName;

(vii) Change data type of decision in Apply to char(1)

Ans alter table apply modify(decision char(1));

(ix) Remove column sizeHs from Student table

Ans alter table student drop column sizeHs;

(x) Drop primary key from college.

Ans alter table college drop primary key.

(xi) Make cName, major unique pairwise such as Stanford CS, Stanford EE

Ans alter table apply add constraint unique apply unique (cName, major);

(xii) Add cName as foreign key in Apply table referencing table college using on delete cascade.

Sol alter table apply add constraints fk_apply foreign key (cName) references college on delete cascade.

(xiii) Modify foreign key on sID in Apply table to foreign key on delete set null.

Sol alter table apply add constraints fk_apply foreign key(sID) references student on delete set null.

(xiv) Rename column enrollment to enroll in College Table.

Ans alter table collegerename column enrollment to enroll;

# Assignment – 3

Q1 Produce a combine table in which each student is combine with every other application

Ans SELECT Students.SID, Apply.SID From Student
INNER JOIN APPLY ON Student.SID = Apply.SID;

Q2 Give Student ID, name, GPA and name of college and major each student applied to

Ans SELECT Student.SID, Student.sName, Student.GPA, Apply.cName, Apply.major FROM Student INNER JOIN Apply ON Student.SID = Apply.SID;

Q3 Find detail of applications who applied to California State.

Ans SELECT Apply.cName, College.cName, College.State
FROM Apply INNER JOIN College.
On Apply.cName = College.cName And College.state = 'CA';

Q4 IDs, name, GPA of students and name of college with GPA > 3.7 Applying to Stanford.

Ans4 Select Student.SID, ~~Student~~ name Student.SID
And student GPA > 3.7 and Apply.cName = 'Stanford';

Q5. Find detail of student who apply to CS major and their application are rejected.

Sol SELECT Student.SID, Students Name, Apply.SID, Apply.major Apply.decision From Student INNER JOIN Apply
On Student.SID = Apply.SID And Apply.major = 'CS'
And Apply.decision = 'N';

6. Find detail of student and application who applied to colleges at New York.

```
SELECT Apply.cName, Apply.major, College.cName,
College.enrollment, College.state From Apply
INNER JOIN College On Apply.cName = College.cName
AND College.state = 'NY';
```

7. Find detail of student who have not applied to any of college.

```
SELECT Student.SID, Student.sName, Student.GPA,
Student.sizeHS From Student LEFT JOIN Apply
ON Student.SID = Apply.SID where Apply.SID IS NULL;
```

8. Find college where no student have applied

```
SELECT cName From College Left JOIN Apply
On College.cname = Apply.cName where Apply.cName
is NULL
```

9. Find SID who have any one application.

```
SELECT SID From Apply GroupBy SID
HAVING count(SID) = 1;
```

10. Find Name and GPA of applicants who apply to any college whose enrollment is not more that 25000.

```
SELECT StudentsName, Student.GPA, Student.SID, Apply.SID,
Apply.cName, College.collegeName, College.enrollment
FROM (Student INNER JOIN Apply On Student.SID =
Apply.SID) INNER JOIN College ON
Apply.cName = College.collegeName And College.enrollment
<25000);
```

11. Find pair of students(SID) having same GPA: (each pair should occur Justonce in result)

```
Select student.SID, X.SID, student.GPA
from student, student X where student.SID != X.SID
and student.GPA = X.GPA;
```

# Assignment - 4

1. Count the total number of students.

```
SELECT count(*) from student;
```

2. Calculate the average GPA of all student

```
SELECT avg(gpa) as Average from student;
```

3. Determine the minimum and maximum GPA. Rename the titles as 'max_GPA' and 'min_GPA' respectively.

```
SELECT max(gpa) as max_GPA, min(gpa) as
min_GPA from student;
```

4. Count the number of students having GPA greater than or equal to 3.7

```
SELECT count(gpa) from student WHERE gpa>=3.7;
```

5. Find Maximum, Average, Minimum, total GPA of all student

```
SELECT max(gpa) avg(gpa), min(gpa), sum(gpa)
from student;
```

6. Find total number of college in our Application Database.

```
SELECT count(distinct cname) from apply;
```

7. Find how many different major a student had applied in.

```
SELECT count(distinct major) from apply;
```

8. Find total no. of Applications in our Application Systems Database.

SELECT count(sid) from apply;

9. Find average of all distinct GPA.

SELECT avg(distinct gpa) from student;

10. Display the total number of application accepted

SELECT count(sid) from apply where decision='Y';

11. Find number of students having GPA > 3.4 and coming from high school having size > 1000.

SELECT count(sid) from student where gpa > 3.4 and sizeHS > 1000;

12. Find how many student applied to 'marine biology'.

SELECT count(sid) from apply where major in ('marine biology');

13. Find how many applications were rejected and accepted by the colleges.

SELECT count(sid), decision from apply group by decision;

14. Find how many student applied to a particular major. (show count(sid) as No. of applications).

SELECT count(sid), major from apply group by major;

# Assignment-5

1. IDs and names of students who have applied to major in CS at some college.

SELECT sid, sname from student where sid in (select sid from apply where major = 'CS');

2. Find ID and name of student having same high school size as Jay.

SELECT sid, sname from student where sizeHS in (select sizeHS from student where sname = 'Jay');

3. Find ID and name of student having same high school size as Jay but result should not include Jay.

select sid, sname from student where sizeHS in (select sizeHS from student where sname = 'Jay') and sname not in ('Jay');

4. Find the name of student with their GPA and sid whose GPA not equal to GPA of Irene

select sid, sname, gpa from student where gpa not in (select gpa from student where sname = 'Irene');

5. Find college where any student having their name started from J have applied?

select cname from apply where sid in (select sid from student where sname like 'J%');

6. Find all different major where Irene has applied?

```
select distinct major from apply where sid in (
  select sid from student where sname = 'Irene');
```

7. Find IDs of student and major who applied in any of major Irene had applied to?

```
select distinct sid, major from apply where
major in (select major from apply where sid in
(select sid from student s1 where sname = 'Irene'));
```

8. Find IDs of student and major who applied in any of major Irene had applied to? But this time exclude Irene SID from the list.

```
select distinct sid, major from apply where
major in (select major from apply where sid in
(select sid from student where sname = 'Irene'))
and sid not in (select sid from student where
sname = 'Irene');
```

9. Give the number of college Jay applied to? (Remember count each college once no matter if he applied to same college twice with different major).

```
select count (distinct cname) from apply where
sid in (select sid from student where sname =
'Jay');
```

10 Find sID of student who applied to more or same number of college where Jay has applied?

select sid from apply where cname in (select cname from apply where sid in (select sid from student where sname = 'Jay'));

11. Find details of students who applied to major CS but not applied to major EE? (SID 987, 876, 543 should only be include in result).

select * from student where sid in (select sid from apply where major = 'CS' MINUS select sid from apply where major = 'EE'));

12. All colleges such that some other college is in same state. (Cornell should not be part of result as no other college in NEW York. Hint: use exists).

select cname from college c1 where exists (select state from college c2 where c1.state = c2.state and c1.cname != c2.cname);

13. Find the college with highest enrollment

select cname, enrollment from college c1 where not exists (select enrollment from college c2 where c1.enrollment < c2.enrollment);

14. Find name of student having lower GPA:

select sname from student s1 where not exists (select sname from student s2 where s1.gpa > s2.gpa);

15. Find the most popular major

```
select major, count(sid) from apply group by major
having count(sid) = (select max(count(sid)) from
apply group by major);
```

16. Find SID, sName, sizeHS of all students NOT from smallest HS

```
select sid, sizehs, sname from student where sizehs! =
(select min(sizehs) from student);
```

17. Find the name of student who applies to all the colleges where SID 987 has applied?

```
select sname, sid from student where not exists (
select distinct cname from apply where
sid = 987 minus select distinct cname from
apply where sid = student.sid);
```

18. Find college where all the student have applied.

```
select cname from college c where not exists ((
select sid from student s) minus (select sid from apply
a where a.cname = c.cname));
```

19. Find sid of student who have not applied to Stanford.

```
select sid from student minus select distinct sid
from apply where cname = 'Stanford';
```

20. Find sid of student that applied to both Stanford and Berkeley.

```
(select sid from apply where cname = 'Stanford')
intersect (select sid from apply where cname =
'Berkeley');
```

21. Give list of all names including all names of colleges and students.

select sname from student union select cname from college;

22. Create a table Application Info having columns SID int, sName: varchar2(10) and number_of_applications: number(2) they filled?

① create table ApplicationInfo(sid int, sname varchar(10), number_of_applications numeric(2));

② insert into applicationInfo select s.sid, s.sname, nvl(a.no_of_application, 0) from ((select sid, sname from student) s left outer join (select sid, count(*) as no_of_application from apply group by sid) a on s.sid = a.sid);

select * from applications;

23. Create table ApplicationData and load with ID, name and college where they applied ~~that~~ with state of college (remember to include details of all students that have applied or not applied) on runtime using single query.

create table ApplicationData as (select distinct sid, cname, state from apply natural join college);

24. Stanford decide not to take any student who have also applied to its rival Berkeley turn their application decision to N

update apply set decision='N' where cname='Stanford' and sid in ((select sid from apply where cname='Stanford') intersect (select sid from apply where cname='Berkeley'));

25. Delete applications that are filled to city 'New York'.

delete from apply where cname in (select cname from college where state='NY');

# Assignment - 6

1. Create a new column DOB in student table.
   Alter table student add DoB date;
2. Insert DoB for each student in corresponding table using above instance of student table.
   Update student SET DOB = '7-APR-95' where SID = 234;
3. Find average of GPA round off to 2 decimal places.
   Select round (avg (gpa), 2) from student;
4. Find year of DoB of student having less than 1000
   Select extract (year from dob) from student where sizeHS < 1000;
5. Compute age of each student.
   Select extract (year from sysdate()) - extract (year from dob) from student;
6. Display name of all students in uppercase and name of college they applied in lower case.
   Select upper (sname), lower (cname) from student s, apply a where s.sid = a.sid;
7. Find fourth alphabet of each student.
   Select substr (sname, 4, 1) from student;
8. Find sID and sName of student whose sName has string length greater than 3
   Select sid, sname from student where length (sname) > 3

9. Find floor, ceiling and truncate (to one decimal place) value of average CGPA.

Select FLOOR(avg(gpa)), ceil(avg(gpa)), TRUNCATE (avg(gpa),1) from student;

10. Display details of all students whose SID is even

Select * from ~~even~~ student where mod(sid,2)=0;

11. Compute Square Root of 900 and 247

```
SELECT sqrt(900), sqrt(power(24,7)) from dual;
```

12. Consider the string "Peter Piper picked a peck of pickled peppers. A peck of pickled peppers Peter Piper picked. If Peter Piper picked a peck of pickled peppers, where the peck of pickled peppers Peter Piper picked?" Find 6th occurence of string 'pick'

```
Select instr ("Peter Piper picked a peck of pickled
Peppers. A peck of pickled peppers Peter Piper
Picked a peck of pickled peppers, where the
peck of pickled peppers Peter Piper picked?','pick',
1,6) as pick from dual;
```

13. Consider String 'Satya Nadella' replace this using the key

```
Select TRANSLATE ('Satya Nadella', 'adelNsty', '
12345670') from dual;
```

14. Display sID, sname and DOB in this format 'February 26, 2014'.

```
Select sid, sname, to_char (dob, 'Month dd, yyyy')
from student;
```